LayerIt! — Actions on the 4 September 2026 draft

Tasks T1–T25, grouped by type. Each row names the part of the thesis it applies to. Section 6 gives the order to do them in.

Some tasks are marked LBD. Those are things you have already written in the ISMIR Late-Breaking Demo submission but that are missing from the dissertation, so the work is mostly moving your own text across rather than writing anything new.

1. Evidence for Chapter 5

Chapter 5 states in four places that a particular measurement was not made. Do these first: they determine how the objectives and research questions have to be worded, and there is no point polishing text that may still change.

ID	Where	What to do
T1	§5.5.2	**Show what happens when the alignment anchors are sparse.** §5.5.2 explains that ScoreWarp interpolates between the onsets given in the MAPS file, so fewer anchors mean less reliable placement, and then says that no controlled comparison was made. The comparison is cheap and it turns that sentence into a result. ● Take the BWV 856 MAPS file you already have and make a reduced copy of it that keeps only the downbeat onsets. ● Regenerate the same figure twice on the same excerpt, once from the full file and once from the reduced one. ● Show the two together and describe where the notation drifts: which passages stay usable and which do not. ● Report this in §5.5.2 in place of the sentence saying the comparison was not run.
T2	§5.4	**Measure what it costs to add a new layer.** §5.4 shows that the layers you already have can be combined and exchanged, then says the cost of adding a genuinely new visualisation type was not quantified. That number is the one claim in the chapter an examiner can verify independently, so it is worth having. ● Implement one layer type that does not exist yet. An onset-novelty curve or a tempo curve would be enough. ● Implement only load_data and draw. Do not modify Visualizer or any existing layer. ● Report the line count of the new class and state explicitly that nothing outside it changed. ● While you are there: your LBD says ten representations are implemented on four shapes, but §5.4 exercises about six. List the full set — that is evidence you already have and are not using. [LBD]
T3	§5.5.1	**Decide whether to test a highly expressive performance.** RQ3 asks under what conditions score-aligned support breaks down. §5.5.1 names expressive playing as one such condition but says no test was run, so the question is raised and not answered. Either close it or stop asking it — both are defensible, but pick one. ● Option A: run one performance with substantial rubato through the whole pipeline (ASAP has candidates that come with the alignment data you need) and show in a figure where the warped score stops being trustworthy.

ID	Where	What to do
		● Option B: keep the current statement, and reword RQ3 in §1.3 so it no longer implies the case was investigated.
T4	§5.2 / §5.3	**Move the beat-tracking analysis from your LBD into the thesis.** In the LBD you show a Clair de Lune composition and then read it: most downbeats are placed correctly, but many false positives are quaver onsets taken as beats, which shows the metrical structure is being misread. You tie that to two conditions known to trouble beat trackers — compound triple metre and highly expressive playing — and point out that both are visible in the figure itself, in the 9/8 signature and the très expressif marking. You then make the argument that each false positive carries a musical address, so the reader can find the passage in the score and hear it in the recording. None of that is in the dissertation. §5.2.1 uses the same recording for audio layers only, with no analysis, and the beat-tracking display sits on BWV 856 with nothing comparable said about it. ● Bring the LBD figure into Chapter 5, in §5.2 or §5.3. ● Write the reading out around it, at more length than the LBD allowed. ● Give it prominence: it is the closest thing you have to a finding rather than a demonstration. [LBD]

2. Claims, structure and content

ID	Where	What to do
T5	§1.3, §5.5	**Re-word the objectives and research questions once T1–T4 are settled.** RQ2 currently reads "How does LayerIt support the creation, modification, and regeneration of score-aligned visualisations through modular layers and scriptable composition?" A question in that form is answered by describing the software, so no result could have come out the other way. Objective 2 has the same shape. Once you know what T1–T4 delivered, the questions can ask for something that could have failed. ● If T2 lands, put the extensibility clause back into RQ2: how readily can new visualisations be added? ● If T1 and T3 land, RQ3 can keep asking under what conditions the support breaks down, with evidence behind it. ● Tag §5.5 as (RQ2, RQ3) rather than (RQ3) alone: it also sets the ceiling on what RQ2 can claim.
T6	§2.3, §4.5	**Introduce the distinction between physical time and abstract time.** Every layer's native time basis is one of two things. Physical time is elapsed duration: audio, feature curves, detected beat positions, probabilities. Abstract time is metrical or structural position: in this thesis, the score. Only abstract-time layers need an alignment step before they can enter a composition; everything else slots straight in at its own scale. That single distinction explains why the score is the only thing ScoreWarp has to touch, and it is currently missing from the thesis. Your LBD §1 already states it: recordings unfold in physical time, symbolic encodings organise it in abstract units, engraved notation renders it in space. ● Introduce it in §2.3, next to the time-warped / link-based pair. ● Carry it into §4.5, where it turns the alignment method from a scaling recipe into a stated condition for admitting a layer. [LBD]
T7	§4.6	**Name the four shapes your layers are built from.**

ID	Where	What to do
		Your LBD says that representations reduce to four shapes — Curve (any function of time), Events (instants), Intervals (time spans) and Field (a dense time–frequency or pitch array) — and that ten representations are implemented on that one interface. Chapter 4 never says this anywhere. It is the clearest available statement of what "composable" actually means in your framework, and it explains why adding a layer is cheap. Put it in §4.6, ahead of the visual, SVG and code levels. [LBD]
T8	§4.6.3 and a new Annex	**Cut the code in the main text down to what is worth reading.** Chapter 4 carries five listings, and some of them show internal workflow rather than anything a reader needs. Code belongs in the body only when a short snippet makes a point that prose cannot. ● Create an Annex and move Listings 4.2, 4.4 and 4.5 into it. ● Keep Listing 4.1 (the MAPS structure) and Listing 4.3 (the Layer interface) in the chapter. ● Add the five-statement script from your LBD §3 in their place. It builds a four-panel composite figure in five lines, which is the most persuasive single thing you can show about the framework. [LBD] ● Renumber every in-text listing reference afterwards.
T9	§4.5	**Shorten the explanation of how layers are scaled.** Your LBD does this in two sentences: two numbers from the score's time axis, its span in seconds and its length in pixels, give the system's only scale factor; nothing else is warped, because each layer converts its own units into seconds and is drawn at that scale. §4.5 takes considerably longer to say the same thing. Use the shorter version. [LBD]
T10	Chapter 4	**Add a diagram of the software.** Chapter 4 has one figure and it is a conceptual box drawing. Since the contribution is an object-oriented design, the reader needs to see the design: Layer and its subclasses, Visualizer, how panels are assembled, and the path from the input files to the final SVG. Note also that the figure numbered 4.3 in the previous version is no longer in the draft. Tell me what happened to it.
T11	§3.4 and the Chapter 5 opening	**Make the evaluation criteria identical in both places.** §3.4 describes five criteria in prose, unnumbered. The opening of Chapter 5 lists four and leaves extensibility out. A reader who checks one against the other finds they disagree. ● Number the criteria in §3.4 as a list. ● Repeat that list, same names and same order, at the head of Chapter 5. ● "Manual-effort reduction" has become "modular composition and regeneration" between the two versions. Decide which you are actually assessing and use it in both places.
T12	§3.4, §1.5, Ch. 5, §5.3	**Choose either "Validation" or "Evaluation" and use it throughout.** Chapter 5 is titled Validation, but §3.4 is "Evaluation Criteria", the chapter's first sentence says it "evaluates", §5.3 is "…Annotation and Evaluation" and §1.5 says "Chapter 5 evaluates the framework". Either word is defensible on its own; the mixture is not. Sweep all four places plus the chapter title.
T13	§2.1.3, §2.1.2, §2.2	**Rebalance Chapter 2.** §2.1.3 explains the waveform, the spectrum, the linear spectrogram, the logarithmic spectrogram, the mel spectrogram, the CQT and the chromagram, with a figure for each. That is textbook material and the reader does not need all of it.

ID	Where	What to do
		Meanwhile §2.2, the section that has to establish the gap your thesis fills, is the shortest in the chapter and discusses two tools. ● Cut §2.1.3 to four figures: waveform, one spectrogram, CQT, chromagram. ● Drop the player-piano history and the notation-format list from §2.1.2. ● Spend the space you recover on §2.2, and make it comparison rather than description.
T14	§3.2	**Split the five prototypes into numbered subsections.** §3.2 runs the five design cycles as continuous prose, so they do not appear in the table of contents and cannot be cross-referenced from elsewhere. Make them §3.2.1–§3.2.5, one per prototype, keeping the goal / built / lesson / decision shape you already use.
T15	Figs. 5.2–5.4, Table 5.2	**Bring the rest of the Chapter 5 figures up to the standard of Figure 5.1.** ● Figures 5.2–5.4 mostly have no axis labels or units, where Figure 5.1 has them on every panel. Add them. ● None of the figures has a legend for the markers (red, black, dashed). Add one, or describe them fully in the caption. ● Figure 5.2's mel spectrogram is washed out beside Figure 5.1's. Regenerate it with the same colormap and normalisation. ● Table 5.2 contains a cell reading "Not assessed". Either assess it or remove the row. ● Table 5.2 says that arbitrary algorithmic curves are "outside the workflow considered" for Piano Precision. Piano Precision is built on Sonic Visualiser and inherits its layer system, so check this claim before an examiner does.
T16	Abstract	**Rewrite the Abstract, last of all.** It still describes a single "validation case" when Chapter 5 now has two case studies; it mentions neither the objectives nor the research questions; and it is written in US spelling immediately above keywords in British spelling. Write it when the thesis is otherwise finished and it can describe what is actually there.

3. Your ISMIR Late-Breaking Demo submission

ID	Where	What to do
T17	§1.3, §6.1, References	**Cite your own paper.** The submission is mentioned twice, in the §1.3 contributions list and again in §6.1, and it is phrased correctly as submitted rather than accepted. But there is no entry for it in the references, so a reader has no way to learn what it is called or to find it. ● Add a bibliographic entry with the full title and both authors, and cite it at both mentions. ● Add "under review at the time of writing". ● §1.3 describes the library as "open-source, pip-installable" and §6.1 drops the pip claim. Use the same wording in both.
T18	§1.3 (footnote)	**Declare that part of this work has already been submitted elsewhere.** The thesis Abstract follows your LBD abstract closely for its first several sentences, and both documents use the same Clair de Lune / Pires case study. Reusing your own material in your own dissertation is entirely legitimate, but it has to be stated once, or a reader who comes across both will wonder why they overlap.

ID	Where	What to do
		• Add a footnote to §1.3 along the lines of: "Parts of Chapters 4 and 5 have been submitted as [ref]."

4. Citations and references

ID	Where	What to do
T19	§2.3, References	**Give the alignment terminology its sources.** §2.3 introduces time-warped and link-based alignment with the phrase "In the literature, it is useful to distinguish…" and then cites nobody. The terms come from specific places and the reader needs to be told where, otherwise they read as your own coinages. • Cite Goebl et al. 2025 at the point where time-warped alignment is defined. • Cite Jiang et al. 2025 at the point where link-based alignment is defined. • Add the Weigl TROMPA deliverable (Multimodal Music Information Alignment v2, TR-D3.5, 2020) as the source of the anchor-point vocabulary. It is already in your LBD bibliography. [LBD]
T20	References	**Complete the entries that are missing information.** • Nakamura and McFee are already complete in your LBD bibliography. Copy them across: ISMIR 2017, pp. 347–353, and the 14th Python in Science Conference, 2015, pp. 18–25. [LBD] • Carabias-Orti still reads "In: (Oct. 2015)", with no venue, no DOI and no page numbers, and the author name is incomplete. • Müller has no DOI, and its title is in sentence case where the rest of the list is in title case. • Entry [1], the MEI introduction page, has no author, no year and no access date. • Consider citing Pugin et al. (ISMIR 2014) for Verovio alongside the Reference Book you quote, as your LBD does.
T21	References	**Remove the access dates from papers.** Ten entries carry an access date, and nearly all of them are conference or journal papers with DOIs. An access date is only needed when you consulted a web page whose content could change. Strip them from every paper. The single entry that should carry one, [1], is the one that does not.

5. Form and mechanics

Individual errors have been fixed one at a time, but the sweeps across the whole document have not been done. Leave these until the text is final, so you are not proofreading passages you may still cut.

ID	Where	What to do
T22	Abstract, Ch. 1, §5.2.1, §5.2.3	**Make the spelling British throughout.** The thesis is at 65 "visualisation" against 16 "visualization", plus "analyze" and "organize". The inconsistency splits cleanly by chapter, which makes it a contained job: the Abstract and the whole of Chapter 1 are in US spelling, Chapters 2–4 are British, and Chapter 5 has two leftovers in §5.2.1 and §5.2.3. §1.1 manages both spellings inside a single sentence. • Sweep the Abstract and Chapter 1 first; almost all of it is there.

ID	Where	What to do
		• Two occurrences are correct and must stay: “Sonic Visualizer” inside the quotation in §2.2, and the Visualizer class name in the code.
T23	as listed	**Specific errors still to fix.** • §1.1: “the how and what is visualised, shapes the conclusions we draw” — the comma separates the subject from its verb, and the sentence is also spliced onto the one before it. This is the second sentence of the thesis. • §2.1.2: “a give note” should be “a given note”. The count in that sentence was corrected to 12 and the typo left in place. • §4.4.2: “@xml.id” should be “@xml:id”. One occurrence remains; the correct form appears elsewhere. • §2.3: “A good example of this comes from [17]” — too conversational for the register of the chapter. • §2.1.2: “we’ll present next a quote” — contraction, and conversational. • §3.2 and Table 3.1: “SV-compatible” abbreviates Sonic Visualiser, which should not be given an acronym, and SV is never defined anywhere. • Figure 5.1 caption: “score aligned onsets” needs the hyphen. §1.2: “under served case” likewise.
T24	throughout	**Consistency passes.** • Library names: Librosa / Matplotlib / Modusa are capitalised in §2.2, §2.4, §3.2 and §4.6.3, and lowercase in §4.2 and §4.3. Use the form the projects use for themselves. • Quotation marks: straight ASCII quotes at six points in §2.1.2–2.1.3, where the rest of the thesis has proper typographic quotes. • List punctuation: the format list in §2.1.2 mixes a colon with a hyphen, and ends one bullet with a period and the next without. • Captions: “Example of audio Spectrum”, “Example of Linear spectrogram”, “Example a Chromagram” (missing “of”) and “LayerIt waveform display an excerpt for BWV 856”. All four repeat in the List of Figures, so regenerate it once they are fixed. • Cross-references: “chapter 3” and “chapter 4” in Chapters 5–6 against “Chapter 4” and “Chapter 5” in Chapters 1–4.
T25	as listed	**Further errors found on this reading.** • its / it’s, three times: twice in §2.1.2 and once in §2.1.3. • §2.1.2: “Thats where Verovio comes in”. • Agreement: “Each of these require different visual approaches” (§1.1) and “this kind of music representations always require” (§2.1.1). • §2.1.1: “sheet music usually does not induce explicit information” — wrong verb. • §2.1.1: “throughout History” and “the western standard” — capitalisation wrong in both directions. • §2.1.2: “the community that develops it(Music Encoding Initiative)” — missing space. • §1.2: two comma splices in one sentence, around “deep learning-based thus, the field…, these are essential…”. • §1.3: “The open-source repository containing example for visualizations developed as part of this work.” • §2.1.3: a bare “[11, 12, 14]” left floating after Figure 2.7 with no sentence attached to it.

ID	Where	What to do
		• The Chapter 4 title breaks across two lines in the contents and in the running headers. • DTW and ML appear in the List of Acronyms but are never used; MIR, OO, SVG and XML are used before they are expanded. • §2.3 ends with “In the next chapter, we will return to this tool in more detail”. The next chapter is Methodology, and ScoreWarp is discussed in Chapter 4.

6. Order of work

Do the evidence first. It decides how the objectives and research questions in T5 have to be worded, and everything else in the thesis points at those.

Phase	Tasks	Why in this order
1. Evidence	T1, T2, T3, T4	The only items needing work at the machine rather than at the keyboard. T1 and T2 are cheap, and each turns a stated gap into a result. T4 recovers an analysis you have already written.
2. Claims	T5	Word the objectives and research questions to match what T1–T4 actually delivered. Everything downstream refers back to them.
3. Move across from the LBD	T6, T7, T8, T9	Four insertions of your own existing text. Low effort, and they raise the level of Chapters 2 and 4 more than anything else on this list.
4. Structure	T10, T11, T12, T13, T14	Diagram, criteria, chapter naming, the Chapter 2 rebalance and the §3.2 subsections. Do T13 before the form pass so you are not proofreading text you are about to cut.
5. Your paper and the references	T17, T18, T19, T20, T21	Self-citation, the dissemination footnote, then the bibliography. Much of T19 and T20 is copying from your own paper.
6. Figures and Abstract	T15, T16	Regenerate the Chapter 5 figures once their content is settled, and write the Abstract last.
7. Form	T22, T23, T24, T25	One concentrated pass at the end, over final text only.